INCH-POUND

MIL-B-16028G
14 August 1989
SUPERSEDING
MIL-B-16028F
9 May 1974

MILITARY SPECIFICATION

BUCKETS, CLAMSHELL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers clamshell buckets.

1.2 Classification. Bucket will be one of the following types and sizes, as specified (see 6.2):

1.2.1 Part or identifying number (PIN) (see 6.1.5)	PIN (Code)
Type I - Light duty rehandling.	- A
Type II - General purpose.	- B
Type III - Heavy-duty excavation.	- C
Type IV - Ore rehandling.	- D

1.2.2 Size. The size of bucket will consist of the following, as specified (see 6.2).

Size 38 - 3/8 cubic yard (cu yd).	- 1
Size 50 - 1/2 cu yd.	- 2
Size 75 - 3/4 cU yd.	- 3
Size 100 - 1 cu yd.	- 4
Size 150 - 1-1/2 cu yd.	- 5

Beneficial comments (recomendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3815

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Size 200 -2 cu yd. - 6
Size 250 - 2-1/2 CU yd. - 7
Size 350 - 3-1/2 cu yd. - 8

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

NIL-T-704 - Treatment and Painting of Materiel.
MIL-C-3580 - Cranes and Crane-Shovels, Truck, Crawler and Wheel Mounted, Full-Revolving and Their Attachments, Packaging of.
MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-T-52176 - Teeth, Surface Ripping; (Points, Shanks, Keepers, and Keeper Locks), Bucket and Dipper, Crane-Shovel.
MIL-C-53039 - Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.

(Unless otherwise indicated, copies of federal and military specifications are available from the Naval Publications and Forms Center, [ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publication. The following document form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which is current on the date of the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 36/A 36M - Specification for Structural Steel.

{Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothlng in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The bucket shall be of the two-line clamshell type, operated by the multiple line block and tackles, or by lever arm type arrangement. The bucket unit shall be complete and ready for use.

3.2 First article. When specified (see 6.2), the contractor shall furnish a bucket for first article inspection and approval (see 4.2.1 and 6.3).

3.3 Standard bucket. components, and accessories. Except as specified in 3.6.1 through 3.6.10, the bucket, components, and accessories shall be the manufacture's standard or optional items which meet or exceed the requirements of this specification.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of Individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term recovered materialism means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.5 Capacities and weights.

3.5.1 Plate line capacity. The plate line capacity shall be defined as the number of cubic feet (cu ft) the bucket will hold when heaped with a fine granular material and the surplus stricken off by drawing a straight edge over the top of the scoop. Plate line capacities shall be not less than that specified in table I.

TABLE I. Plate line capacities.

	Size							
	38	50	75	100	150	200	250	350
Plate line capacity (cu ft)	10.0	13.5	19.0	26.5	39.0	52.0	64.0	85.5

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3.5.2 Weights of buckets and counterweights. The weight of the bucket without teeth shall be not less than that specified in table II. The weight of the counterweights shall not exceed the given percentage of the total weight of bucket and counterweights shown in table II for the various sizes of types II, III, and IV buckets.

TABLE II. Bucket weight.

	Size							
	38	50	75	100	150	200	250	350
Type I, weight of bucket only, min (lb)	1,100	1,500	2,300	2,800	4,000	5,100	6,200	7,800
Type II, weight of bucket and counterweights, min (lb)	1,500	2,500	3,000	3,700	5,550	6,650	7,950	9,550
Type II, max percent counter-weights of total weight (percent)	21	17	14	13	12	10	10	10
Type III, weight of bucket and counterweights, min (lb)	1,900	2,750	3,800	4,450	6,150	7,650	9,150	10,450
Type III, max percent counter-weights of total weight (percent)	18	14	12	12	12	12	11	9
Type IV, weight of bucket and counterweights, min (lb)				6,000	8,600	10,200	12,700	15,000
Type IV, max percent counter-weights of total weight (percent)				13	12	12	10	8

3.6 Components.

3.6.1 Bucket shell. The bucket shell shall consist of two scoops hinged together. Tagline attachments shall be furnished. The scoops shall be designed in accordance with best commercial practice to handle efficiently the material for which the bucket is intended. The inside surfaces shall be smooth and free from rivet or bolt heads and other projections. The scoop at

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the corner brackets shall be of sufficient cross-section and strength to pemit full transference of maximum design stress to the corner arms. Scoop plate material shall be steel which meets or exceeds the requirements of ASTM Designation A 36. Scoop plates shall be joined by either riveting or welding unless of the one-piece type.

3.6.2 Lips. The lips shall form the cutting edge of the scoops, and shall extend in one piece across the bottom and up both sides to the top of the scoop . The lips shall be of adequate width and thickness to provide for secure attachment of the teeth. Lips shall be of abrasion-resistant alloy steel of the type that has been proven satifactory in extensive use, and shall be heat treated to not less than 185 Brinell hardness. The design of all lips or cutting edges shall be such as to assure proper fit, eliminate resistance for material entering the scoop, and permit replacement by welding or riveting.

3.6.3 Teeth. Unless otherwise specified (see 6.2), manufacturer's standard bucket teeth shall be furnished. When specified (see 6.2), teeth, including points, shanks, keepers, and keeper locks, shall conform to MIL-T-52176, as applicable. Types II, III, and IV buckets, and when specified (see 6.2), type I bucket shall be provided with not less than five bottom teeth. When specified (see 6.2), manufacturer's standard side and corner teeth shall be furnished. The teeth shall be readily replaceable by the use of standard tools. The shanks shall be attached to the bucket with plow bolts fabricated from steel having a minimum tensile strength of not less than 105,000 pounds-force per square inch.

3.6.4 Hinge. The hinge shall be of steel, either of the single or double shaft type, and arranged for multiple line or lever arm type closing action. In the multiple line block and tackle type of closing action, the upper block shall be attached to the bucket head and the lower block attached to the hinge body or the lever arm. In the lever arm type of closing action, the lever arm or arms shall be either integral with or rigidly secured to the hinge body or to one or both of the scoops. Where the single shaft design is used, the scoops that rotate on the shaft shall be equipped with replaceable bushings. Where the double shaft design is used, both scoops shall be pinned to the shafts which, in turn, shall rotate on replaceable bushings. Provision shall be made for attaching counterweights in accordance with good commercial practice. The reeving of the closing line shall avoid reverse bends over the sheaves. Stops shall be provided for the correct digging position of the scoops.

3.6.5 Bucket head. The bucket head shall be a steel casting or weldment with attachments for the corner arms, closing line sheaves, and sockets for the hoisting and, when necessary, the closing line.

3.6.6 Corner arms. Corner arm shall be one-piece forged steel, or of bar steel with welded or upset bosses.

3.6.7 Brackets. The hinge and arm brackets shall be of steel and of sufficient size and strength to accommodate stresses caused by lifting, dropping, bending, or twisting forces.

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3.6.8 Sheaves. Unless otherwise specified (see 6.2), sheaves shall be provided with manufacturer's current standard bearings or bushings. Sheaves shall have smooth grooves and a pitch diameter of not less than 15 times the diameter of the largest wire rope recommended by the bucket manufacturer for use with the bucket. When specified (see 6.2), sheave groove cleaners shall be furnished.

3.6.9 Rollers, Shafts, and pins. Rollers. shafts, and pins shall be the manufacturer's current standard.

3.6.10 Bushings. Bushings shall be heat-treated alloy steel, alloy bronze, or cast iron, as applicable.

3.7 Cleaning, treatment, and painting. Surfaces nomally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as soil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean. dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags. orange peel, or other defects. When specified, (see 6.2), the portions of the (item} normally painted internally and externally shall be cleaned, treated and painted in accordance with MIL-T-704, type F or G, as applicable. Top coat color shall be green 383 conforming to MIL-C-46168 or MIL-C-53039 as applicable.

3.8 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less than l/8-inch in diameter shall be used to affix the plates. Nomenclature contained in contract item description shall be placed in the top blank.

3.9 Lubrication. Lubrication fittings shall be provided in accessible, protected positions for bearrings and moving parts requiring lubrication.

3.10 Workmanship. All parts, components, and assemblies of the bucket, including castings, forgings, molded parts, stamping, bearings, seals, machined surfaces, and welded parts, shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free from burrs, sharp edges, and corners are required.

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3.10.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.10.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.10.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.10.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.10.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with

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all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. The supplier is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Examination. Each bucket shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the supplier's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.2.1 First article inspection. First article inspection shall be performed on one bucket when a first article sample is required (see 3.2 and 6.2). This inspection shall include the examination of 4.2 and the test of 4.3. The first article may be a standard production item from the supplier's current inventory provided the bucket meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining buckets to be furnished under the contract.

4.3 First article tests. When first article is required, the first article bucket of each type and size shall be subjected to the tests specified in 4.3.1 through 4.3.3 as applicable. The first article tests shall be performed by the supplier. The supplier shall provide all material and equipment to perform all testing specified. Failure to pass any phase of the required tests shall be cause for the Government to refuse acceptance of all buckets until corrective action has been taken.

4.3.1 Materials rehandling tests. Type I bucket procured without teeth shall be subjected to materials rehandling tests. Material for testing shall be loose or stockpiled earth, sand, gravel, or other comparable material. The open bucket shall be set down on the material, closed without interruption, hoisted, swung to the side, and dumped. The average load from 10 cycles shall be at least equal to plate line capacity of the bucket.

4.3.2 Digging tests.

4.3.2.1 Type III bucket. Type III bucket shall be subjected to digging tests. Material for testing shall be in-place dry clay or soft shale. The surface of the test material shall be relatively smooth with not over 2 percent average grade. he open bucket shall be set down on the material. Power shall be applied to the closing line until the bucket is closed.

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Closing shall be positive and uninterrupted. The bucket shall be raised and conveyed to the proper location where material shall be completely dumped. The average load from 10 cycles shall be at least equal to plate line capacity of the bucket.

4.3.2.2 Type IV bucket. Type IV bucket shall be subjected to tests specified in 4.3.2.1 except that the test material shall be loose or stockpiled earth, rock, or a combination.

4.3.3 Combination tests. Type I bucket procured with teeth and type II bucket shall be subjected to the following tests:

a. With teeth installed, the bucket shall be tested as specified in 4.3.2.1, except that the test material shall be moist clay or moist loam.
b. With teeth removed, the bucket shall be tested as specified in 4.3.1.

4.4 Production unit tests. The contractor's testing system shall, as a minimum, assure that the bucket is capable of meeting the requirements of this specification.

4.5 Packaging inspection. The packaging inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-C-3580.

5 PACKAGING

5.1 Preservation. packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-C-3580 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers the standard types of clamshell buckets to be used for earth excavation and handling of loose materials.

6.1.1 Type I, light duty rehandling. This type is intended for handling all kinds of loose materials, including concrete aggregate. Limiting size for crushed rock or gravel is approximately 2-1/2 inches.

6.1.2 Type II, general purpose. This type is intended for excavating moist clay, loam, sand, and loose gravel. By removing the teeth, the bucket may be used for rehandling materials.

6.1.3 Type III, heavy-duty excavation. This type is intended for excavating dry clay, gravel, or soft shale.

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6.1.4 Type IV, ore rehandling. This type is intended for rehandling and stockpiling ore.

6.1.5 Part or Identifying numbers (PIN). The PINs were developed to identify items covered by this specification for cataloging purposes. The PIN consist of this specification identifier (M16028), the type and size as determined by classification section 1 and type and size as per table I.

Example - Heavy Duty Excavation 1 cu. yd.

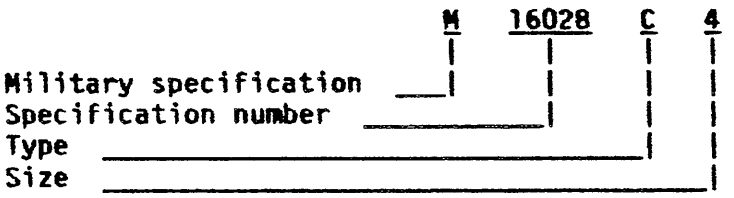


TABLE II. Bucket Type and Size.

Type	Size							
	1	2	3	4	5	6	7	8
A								
B								
C				X				
D								

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and size of bucket required (see 1.2 and 1.2.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article is required for inspection and approval (see 3.2, 4.2.1 and 6.3).
e. When teeth, points, shanks, keepers, and keeper locks shall conform to M1L-T-52176 (see 3.6.3).
f. When type I bucket with teeth is required (saee 3.6.3).
g. When side and corner teeth are required (see 3.6.3).
h. When sheave bearings other than specified are required (see 3.6.8).
i. When sheave groove cleaners are required (see 3.6.8).
j. When cleaning, treatment and painting in accordance with

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MIL-T-704, type A is required (see 3.7).
k. For Army procurements specify MIL-T-704, type F or G paint (see 3.7).
l. Finish coat color required (see 3.7).
m. Level of preservation and packing required (see 5.1).

6.3 First article. When a first article is required, it should be tested and approved under the appropriate provision of paragraph 7-104.55 of the Armed Services Procurement Regulations. The first article should be a first production item consisting of one complete bucket unit or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in all procurement Instruments, regarding arrangements for examinations, test and approval of the first article.

6.4 Subject term (key word) listing.

Cable Operated.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD
Air Force - 84

Preparing activity:
Navy - YD

(Project 3815-0099)

Review Activity:
DLA - CS

User Activity:
Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-16028G	Buckets, Clamshell

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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